Fatemeh Shirvani

M.Sc. student — Interaction, Graphics & Design · Real-time rendering, simulation, XR, machine learning

ftm.shirvani@gmail.com
linkedin.com/in/fatemeh--shirvani
github.com/FatemehShirvani
fatemehshirvani.github.io · Paris, France

EDUCATION

M.Sc. Interaction, Graphics & Design
2025 — present

Institut Polytechnique de Paris

Real-time rendering, physically based simulation, geometry processing, mixed reality and HCI.

B.Eng. Computer Engineering — Intelligent Systems

Isfahan University of Technology

Foundations in algorithms, machine learning and signal processing.

EXPERIENCE

R&D Intern — 3D Gaussian Splatting for Immersive VR
Jun — Aug 2026

EssilorLuxottica

- Recreated a physical refraction room as a room-scale VR scene: photographic capture and preprocessing, Structure-from-Motion calibration with COLMAP, and 3D Gaussian Splatting training.
- Evaluated held-out novel views with image-quality and perceptual metrics; integrated the scene in Unity with VRSplat and OpenXR to test stereo rendering, 6-DoF navigation and runtime performance.
- Identified reconstruction limits around low-texture surfaces, reflections, thin structures and disocclusions.

3DGS · VRSplat · Unity · COLMAP · OpenXR

Research — Motor Learning from Foot-Drawn Shapes
Jun — Sep 2024

- Studied how practice changes movement quality over a seven-session foot-drawing task; corrected sampling and quantization artifacts.
- Extracted structural and kinematic features (SSIM, duration, velocity, curvature, jerk, smoothness) and compared clustering and classification models.
- Leave-one-subject-out evaluation: 96% shape-specific accuracy, 79% pooled Random Forest, 81% early-vs-late session grouping on unseen participants.

Python · MATLAB · Signal Processing · Machine Learning

SELECTED PROJECTS

Variance Soft Shadow Mapping
2026

WebGPU renderer comparing PCF, PCSS, VSM, SAVSM and a hybrid VSSM on thin foliage.

WebGPU · WGSL

Plant Growth Modeling
2026

Procedural tree growth, seasons and wind in real-time C++/OpenGL with L-systems.

C++ · OpenGL

Smoothed Particle Hydrodynamics
2026

2D SPH solver for incompressible flow around moving obstacles.

C++ · Numerical Methods

Ski Locomotion in Virtual Reality
2026

Embodied Meta Quest locomotion that turns ski-pole strokes into motion, evaluated with timed tasks.

Unity · C# · Meta Quest

Loop Subdivision Surfaces
2026

Progressive mesh smoothing with interior/boundary rules, wireframe inspection and undo.

C++ · OpenGL · Geometry Processing

Multi-Body Rigid Body Simulation
2026

Boxes and cylinders under impulses, Coulomb friction, restitution and sleep states.

C++ · Rigid Body Dynamics

Influencer Detection on French Twitter
2025

LoRA-tuned CamemBERT + behavioral features + XGBoost — 85.7% accuracy.

CamemBERT · LoRA · XGBoost · NLP

Interactive Wardrobe Customization
2025

Playful 2D fashion game exploring direct-manipulation techniques.

Unity · C# · Interaction Design

SKILLS

Graphics & Rendering	C++, OpenGL, WebGPU, GLSL / WGSL, 3D Gaussian Splatting, COLMAP
XR & Interaction	Unity, C#, OpenXR, Meta Quest, React
Machine Learning & Data	Python, PyTorch, scikit-learn, XGBoost, OpenCV, NumPy, MATLAB
Tools	LaTeX